Relational Actualism:

The Engine of Becoming

Joshua F. Sandeman

Independent Researcher, Salem, Oregon

March 2026

A synthesis of the Relational Actualism research program, tracing the logical cascade from a single ontological commitment through quantum mechanics, gravity and cosmology, and the emergence of biological life and intelligence. The program comprises six companion papers (RAQM, RAGC, RACI, RAQI, RAHC [Sandeman, 2026e], RADM [Sandeman, 2026f]); the Monograph is organised in six parts; key theorems have been formally verified in Lean 4/Mathlib. For technical detail and precise status labels on each result, consult the individual papers.

Relational Actualism demonstrates that the universe is fundamentally processural and causal. By showing that time, causality, possibilities, and actualities are interrelated ontological primitives, it generates sweeping consequences that create a coherent predictive framework — resolving many outstanding problems in fundamental physics, cosmology, and complexity science, while offering clear, testable predictions and well-defined open problems for future research.

Contents

Preface: The Engine of Becoming **1**

1 Part I: The Ontological Commitment **1**
1.1 Three Kinds of Property . . . 1
1.2 The Actualization Criterion . . . 2
1.3 Relativistic Consistency and the No-Signaling Theorem . . . 3
1.4 Empirical Predictions of the Quantum Foundation . . . 3

2 Part II: Geometry as Accumulated History **4**
2.1 Replacing the Manifold with a Growing Graph . . . 4
2.2 Conservation as a Local Ledger . . . 4
2.3 Vacuum Energy Suppression (Theorem 1) . . . 4
2.4 The Ledger Partition Theorem (Theorem 2, Lean-Verified) . . . 5
2.5 Intrinsic Expansion and the Four Cosmological Predictions . . . 6

3 Part III: Life as Accumulated Causal History **7**
3.1 The Causal Firewall . . . 7
3.2 The Four Empirical Predictions of RACI . . . 7
3.3 The Assembly Index Correspondence (Theorem 3) . . . 8
3.4 The Markov Blanket Isomorphism (Theorems 4–5, Lean-Verified) . . . 8

4 Part IV: The Information-Theoretic Turn **9**
4.1 The Kinematic Snap as a Physical Decoherence Criterion . . . 9
4.2 The Kinematic Coherence Bound . . . 9
4.3 Quantum Thermodynamics Grounded in the Causal DAG . . . 10
4.4 The Quantum Relative Entropy Connection . . . 10

5 Part V: The Hierarchy of Complexity **11**
5.1 The Problem of Emergence . . . 11
5.2 Emergence as Topological Closure . . . 11
5.3 The Recursive Coarse-Graining Hierarchy . . . 11
5.4 The Mathematics of Graph Steering . . . 12
5.5 The Thermodynamic Cost of Thought . . . 12
5.6 ACD and the End of Epistemic Reductionism . . . 13

6 Part VI: The Mathematical Frontier **13**

Conclusion: One Commitment, Sixteen Consequences **15**

A The Pragmatic Turn: Tractable Tools for a Discrete Ontology **17**

Declarations **19**

Preface: The Engine of Becoming

There is a deep, unacknowledged ghost haunting the halls of modern theoretical physics. It is the ghost of the timeless universe.

For over a century, our most successful physical theories have been written in a mathematical language that possesses no concept of "now." In the equations of classical mechanics, general relativity, and unitary quantum field theory, time is merely a spatialized parameter. The domain and codomain are completely specified, forever. In this mathematical Platonism, the universe simply *is*. But the universe is not a static block. It is an engine of becoming.

Standard QFT is complicit in this failure. Because QFT lacks a formal mathematical mechanism to represent the mere fact that things happen, it fails to distinguish between what is mathematically possible and what is physically actual. It grants full causal and gravitational weight to infinite seas of unactualized virtual potentia—directly producing the $\sim 10^{120}$ discrepancy of the vacuum energy problem.

In reality, the universe does not grant causal power to mere possibilities. The causal structure of the universe is built exclusively through specific, physical mechanisms of becoming: objective on-shell boson exchange and causal severance. Relational Actualism (RA) proposes that time and causality are ontologically primitive—the ground floor of nature, not its emergent ceiling.

From this single commitment, a cascade follows. The measurement problem dissolves. The vacuum energy catastrophe disappears structurally. The Hubble Tension becomes a geometric consequence rather than a mystery. Event horizons become topological severances of a growing causal graph. Biological life becomes the inevitable product of a universe that preserves its history. Intelligence becomes the physical process of learning to steer that history. The Boltzmann Brain becomes not merely improbable but categorically impossible.

This monograph traces that cascade in full. It is not a collection of separate results; it is a single argument, unfolding from a single premise.

A note on scope and formal verification. The formal developments span established results, partial derivations, and conjectures at different levels of rigor. The companion papers (RAQM, RAGC, RACI, RAQI, RAHC [Sandeman, 2026e], RADM [Sandeman, 2026f]) are explicit about the precise status of each claim. The Ledger Partition Theorem (Part II) and the Markov Blanket Isomorphism and Epistemic-Physical Bound (Part III) have been formally verified in Lean 4/Mathlib, establishing the logical soundness of their discrete mathematical cores. RAQM has been submitted to *Foundations of Physics* and cleared desk review as of March 2026. Readers seeking technical apparatus should consult the individual papers; what follows is the architecture of the whole.

1 Part I: The Ontological Commitment

1.1 Three Kinds of Property

The first step is a taxonomy. Physical properties divide into three kinds, and the distinction between them is not merely philosophical—it is load-bearing for the entire program.

Invariant properties have exactly one possible state. The charge of an electron is invariant: it is always $-e$. No interaction can change this.

Classical properties vary, but deterministically. Given a complete specification of the relevant boundary conditions, the future state of a classical property is fixed. The trajectory of a planet is classical: genuinely uncertain only to an observer lacking sufficient information, never genuinely open to nature.

Intrinsically relational properties cannot assume definite values outside a specific context of causal interaction. The spin of a newly created particle, entangled with its twin, is intrinsically relational: it is not merely unknown but genuinely indefinite until actualized by an appropriate

interaction. This is not a statement about our ignorance. It is a statement about the structure of the world. Quantum randomness is not accidental but necessary—a logical consequence of what relational properties are.

Crucially, this is not a statement about consciousness or observation. It is a statement about *causal interaction*. The boundary between the classical and the quantum is not the boundary between the macroscopic and the microscopic; it is the boundary between systems that have been fully actualized by their environment and systems that have not.

1.2 The Actualization Criterion

An interaction constitutes an *actualization event* when it produces an irreversible increase in the entanglement entropy of the system with its asymptotic environment, generating a robust and redundant causal record recoverable in principle by any future observer in the causal future of the interaction.

This criterion is:

- **Basis-independent.** Entanglement entropy $S(A) = -\mathrm{Tr}(\rho_A \log \rho_A)$ is defined at the level of the density matrix, not a perturbative expansion. It does not depend on a choice of computational basis.
- **Non-perturbative.** It applies in strongly coupled regimes—QCD confinement, condensed matter critical points—wherever the underlying dynamics produce an irreversible environmental entanglement.
- **Free of new constants.** Unlike dynamical collapse models (GRW, CSL, Diósi-Penrose [Bassi et al., 2013, Diósi, 1989, Penrose, 1996]), the criterion introduces no new fundamental constant. The threshold is set entirely by the existing physical interaction.
- **Sufficient in all regimes.** Whether necessity holds in full generality—beyond Markovian open systems and perturbative QFT, where necessity can be made explicit—is an open refinement requiring an algebraic-QFT formulation via relative entropy monotones on local von Neumann algebras.

In weakly coupled QFT, the exchange of a real, on-shell excitation satisfying $p^\mu p_\mu = m^2c^2$ is a sufficient condition for actualization. Off-shell virtual exchanges—mediating the Casimir effect, van der Waals forces, the Lamb shift—leave the environment in its ground state and produce no persistent causal record. The causal DAG is built from S-matrix poles, not from the full off-shell generating functional.

The Kinematic Snap: the microscopic mechanism of irreversibility. The precise physical mechanism that converts reversible potentia into irreversible actuality can be stated in standard QFT terms. When two particles interact, their wavefunctions overlap via virtual, off-shell bosons ($p^\mu p_\mu \neq m^2c^2$). Virtual bosons cannot propagate to the asymptotic environment, so they cannot balance the Local Ledger and write no vertex to the DAG. The interaction remains reversible for as long as only virtual exchange occurs.

Actualization occurs at the *kinematic threshold*: the moment the interaction possesses sufficient energy-momentum to elevate a mediating boson to the mass shell ($p^\mu p_\mu = m^2c^2$). An on-shell boson propagates to infinity; the LLC demands permanent recording; an irreversible vertex is written. The reversible hum of virtual potentia snaps into a discrete, permanent causal fact. Irreversibility is kinematic, not statistical. The threshold is a binary condition on the dispersion relation, not a thermodynamic average. No observer is required: hydrogen forms when the on-shell photon carrying the binding energy (13.6 eV) propagates away, regardless of whether anyone is watching. Finite decoherence time is explained by the physical propagation time of the emitted on-shell bosons to the environmental boundary.

The Unruh vacuum and the stationarity criterion. In the correct algebraic QFT language, the Rindler observer sees the *restriction of the Minkowski vacuum to the Rindler wedge algebra*—a KMS (thermal equilibrium) state with respect to boost automorphisms, per the Bisognano-Wichmann theorem. This KMS state is in equilibrium with respect to modular flow: its relative entropy with respect to the vacuum reference state is constant ($\Delta S_{\text{rel}} = 0$). No irreversible increase occurs; no net on-shell boson emission into the asymptotic environment takes place. The KMS state is the universe's stationary potentia as seen from an accelerated frame—it is not generating actualization events. Both observers therefore agree: in the Minkowski vacuum, no actualization is occurring. The apparent disagreement on particle content is a modal decomposition artefact, not a disagreement about causal facts. At the extreme limit—by the Einstein Equivalence Principle, both extreme acceleration and extreme mass-density are the same topological phenomenon—the local actualization density saturates $R_{\max}$, the graph shears, and a causal severance occurs. This is the RA account of what standard GR calls a singularity: not an infinity, but a graph cut obeying the Ledger Partition Theorem.

Frame-independence of the actualization criterion. The criterion is Lorentz-covariant. Relative entropy $S(\rho\|\sigma)$ is a diffeomorphism-invariant scalar. A strictly irreversible increase in relative entropy with respect to the vacuum state is an objective, coordinate-independent fact. The Kinematic Snap operates at all energy scales — from the meV binding energies of molecular bonds to the TeV interactions at colliders — and is observer-independent at every scale. The on-shell particle language is the clean limiting expression of this criterion in weakly coupled regimes; the fundamental criterion is the relative entropy increase, which is covariant in all regimes.

1.3 Relativistic Consistency and the No-Signaling Theorem

A globally consistent wave function updated by local actualization events seems to require a preferred universal "now"—a violation of Lorentz invariance. Relational Actualism resolves this by formalizing the global state as a functional $\Psi[\sigma]$ on an arbitrary spacelike hypersurface σ via the Tomonaga-Schwinger equation [Tomonaga, 1946, Schwinger, 1948]. The state is updated along any valid spacelike hypersurface through an actualization vertex; the resulting observable statistics are foliation-independent.

This is not assumed but derived. The RAQM consistency condition establishes that for any operator $\hat{O}_B$ supported in a region B spacelike-separated from an actualization event at x:

$$\mathrm{Tr}\left[\hat{O}_B \sum_k J_k(\rho)\right] = \mathrm{Tr}[\hat{O}_B \rho], \tag{1}$$

where $\{J_k\}$ is the localized quantum instrument at x. The non-selective (outcome-averaged) actualization map leaves expectation values at spacelike separation invariant. Superluminal signaling is rigorously forbidden. Conditioned (post-selected) branches can differ at spacelike separation, but the specific outcome is not locally controllable from B—no information is transmitted.

1.4 Empirical Predictions of the Quantum Foundation

The RAQM framework makes three predictions that distinguish it from all competing interpretations [Sandeman, 2026a]:

(P1) Finite actualization propagation velocity. The actualization front—the boundary between superposed and actualized regions—propagates at a finite velocity bounded by the Lieb-Robinson speed $v_{\text{LR}} \sim J\ell/\hbar$ in lattice systems. A distant site B following an actualization event at site A should exhibit a time delay $\Delta t \geq |r_A - r_B|/v_{\text{LR}}$ before coherence collapses. Instantaneous collapse models predict no such delay.

(P2) Environment-dependent transition duration. The actualization timescale scales as $\tau_{\text{act}} \sim \hbar/\Delta E_{\text{env}}$, where ΔE_{env} is the characteristic environmental coupling energy. Systematically varying

environmental coupling in superconducting circuit experiments should produce a proportional variation in transition duration—a clean quantitative test, consistent with Minev et al. [Minev et al., 2019] and Pazourek et al. [Pazourek et al., 2015].

(P3) A sharp reversibility threshold. Below the actualization threshold (insufficient environmental entanglement), quantum transitions are reversible in principle. Above it, the causal record is permanent and cannot be erased regardless of intervention. This threshold is not present in decoherence-only models, which deny that a sharp actualization event occurs at all.

2 Part II: Geometry as Accumulated History

2.1 Replacing the Manifold with a Growing Graph

If time is primitive and spacetime is actively forged by actualization events, the continuous differentiable manifold of General Relativity is the wrong mathematical object. It spatialized time before we began; we need a mathematics of process. Relational Actualism replaces the manifold with a Directed Acyclic Graph (DAG) $G = (V, E)$, extending Causal Set Theory [Dowker, 2005, Sorkin, 2003, Rideout and Sorkin, 2000]. Each vertex $v \in V$ is an actualization event; each directed edge $(u, v) \in E$ is the causal propagation of physical state from u to v. The graph grows monotonically: the future is not already there, it is being written.

The DAG has a fundamental *bandwidth limit* $R_{\max}$—a maximum rate at which the local spacetime geometry can process quantum state transitions. This is not an assumption grafted onto the framework; it follows from combining the holographic bound with the actualization criterion. When the local actualization flux approaches $R_{\max}$, the causal structure saturates and a topological severance occurs.

2.2 Conservation as a Local Ledger

In standard physics, conservation laws are global constraints: the total charge of the universe is constant. In a growing DAG, global constraints are ill-defined because the universe's vertex set is not fixed. Relational Actualism reconceives conservation as a *local ledger condition*: at every vertex $v \in V$, the sum of incoming charges equals the sum of outgoing charges:

$$\sum_{\substack{u \in V \\ (u,v) \in E}} \mathbf{q}(u) = \sum_{\substack{w \in V \\ (v,w) \in E}} \mathbf{q}(w). \tag{2}$$

In the continuum limit $\lambda \to \infty$ of a dense actualization graph, the Local Ledger Condition (LLC) converges to the standard Noether continuity equation $\nabla_\mu J^\mu = 0$. The LLC is more fundamental: it holds in the discrete regime where the manifold approximation breaks down entirely.

2.3 Vacuum Energy Suppression (Theorem 1)

The vacuum energy problem is often described as the worst fine-tuning in all of physics: standard QFT predicts a vacuum contribution $\langle 0|T_{\mu\nu}|0\rangle \approx M_{\rm Pl}^4 \sim 10^{74}$ GeV^4 to the metric source, while observations require $\rho_\Lambda \sim 10^{-47}$ GeV^4—a discrepancy of $\sim 10^{120}$. Standard renormalization subtracts this by hand, requiring extraordinary fine-tuning [Weinberg, 1989].

Relational Actualism dissolves the problem structurally. Define the Actualization Projector $\mathscr{P}_{\rm act}$ as the spectral projection onto the on-shell subspace $\mathscr{H}_{\rm on} = \{|\psi\rangle : p^\mu p_\mu |\psi\rangle = m^2c^2|\psi\rangle\}$, and the RA-modified stress-energy operator as $T^{\mu\nu}_{\rm RA} := \mathscr{P}_{\rm act} T^{\mu\nu} \mathscr{P}_{\rm act}$. The modified Einstein equation is then:

$$G_{\mu\nu} = 8\pi G \langle \psi_{\rm phys}| T^{\mu\nu}_{\rm RA} |\psi_{\rm phys}\rangle. \tag{3}$$

Theorem 1 (Vacuum Energy Suppression, RAGC Theorem 1)**.** $\langle 0| T^{\mu\nu}_{\rm RA} |0\rangle = 0$.

The proof is direct: $\mathscr{P}_{\text{act}}|0\rangle = 0$ because the Fock vacuum contains no on-shell quanta. This is not a renormalization. It is a structural consequence of the RA postulate that the metric is sourced only by actualized states. The vacuum is a superposition of unactualized potentia; $\mathscr{P}_{\text{act}}$ removes it from the gravitational source by definition.

Four mathematical derivations remain before $\mathscr{P}_{\text{act}}$ constitutes a complete gravitational theory: a non-perturbative definition in strongly coupled regimes; agreement with renormalized $T^{\mu\nu}$ including binding energy corrections and loop corrections; a back-reaction fixed-point proof; and Bianchi compatibility. The Bianchi problem has been substantially resolved, as follows.

The vacuum-subtraction prescription and Casimir benchmark. The RA prescription for the stress-energy source is:

$$\langle T^{\mu\nu}_{\text{RA}}\rangle[g,\Phi] \;:=\; \langle T^{\mu\nu}_{\text{ren}}\rangle[g,\Phi] \;-\; \langle T^{\mu\nu}_{\text{ren}}\rangle[g,|0\rangle]. \tag{4}$$

This subtracts the absolute vacuum reference (the configuration-independent $\sim M_{\text{Pl}}^4$ term) while preserving all physically meaningful energy differences that arise from actualized boundary conditions. The Casimir effect is precisely such a difference: the plates are actualized objects that modify the causal graph, and their effect on the field is physically real.

Under this prescription: (a) $\langle 0|T^{\mu\nu}_{\text{RA}}|0\rangle = 0$ by self-subtraction; (b) $\nabla_\mu\langle T^{\mu\nu}_{\text{RA}}\rangle = 0$ by linearity and the Wald axioms; (c) Casimir stress-energy is reproduced exactly; (d) the modified Einstein equation $G_{\mu\nu} = 8\pi G\langle T^{\mu\nu}_{\text{RA}}\rangle$ is Bianchi-compatible. A numerical benchmark at $d = 1\,\mu$m gives Casimir pressure -1.30×10^{-3} Pa, matching experiment.

Scope: Step (b) uses linearity of ∇_μ and conservation of each term under the Wald axioms (established for free fields on globally hyperbolic spacetimes). The flat-space benchmark verifies ordinary divergence ∂_μ rather than the covariant divergence ∇_μ in a non-trivial metric. Demonstrating $\nabla_\mu\langle T^{\mu\nu}_{\text{RA}}\rangle = 0$ on a curved background constitutes the decisive remaining test for D3 (see the Mathematical Frontier).

The emergent metric defence. A deeper resolution: the Einstein equations are not fundamental in RA but are the thermodynamic equation of state for the causal graph, in the same sense as Jacobson's thermodynamic derivation of GR [Jacobson, 1995]. The LLC enforces conservation exactly at every discrete vertex. The continuous Bianchi identity is the continuum shadow of the LLC, not an independent constraint. Remaining open scope: the non-perturbative regime and full curved-spacetime back-reaction — precisely the same open problems as in standard semiclassical gravity.

2.4 The Ledger Partition Theorem (Theorem 2, Lean-Verified)

An event horizon in the RA framework is a maximal causal cut C^*—a set of edges whose removal disconnects the DAG into a causally inaccessible interior and accessible exterior, with all cut edges running from past to future. What happens to conservation across such a cut?

Theorem 2 (Ledger Partition, RAGC Theorem 2 — **formally verified in Lean 4/Mathlib**). *For any causal severance of the spacetime DAG, the Local Ledger Condition holds independently on each disconnected subgraph. The total boundary flux across the cut is zero.*

This theorem has been formally verified in the Lean 4 theorem prover using Mathlib. Its discrete graph-theoretic core—that LLC at every interior vertex implies zero net boundary flux—is logically watertight. The proof strategy proceeds by showing that summing the LLC over all vertices in V_L causes internal edges to cancel exactly, leaving only the boundary flux, which must therefore be zero.

The result dissolves the energy bookkeeping version of the Hawking radiation problem. The question "is $Q_{\text{interior}} + Q_{\text{exterior}}$ conserved?" is not well-posed after a causal severance: Q_{interior} is

inaccessible to the exterior subgraph by the definition of an event horizon. Each side conserves independently. This is not a violation of conservation but a clarification of what conservation means in a growing causal graph. The unitarity question—whether quantum information is preserved across the horizon—is a separate, harder problem requiring the non-perturbative treatment of $\mathscr{P}_{\text{act}}$.

2.5 Intrinsic Expansion and the Four Cosmological Predictions

The growing DAG expands intrinsically: each new actualization event adds vertices and edges, continuously weaving new causal structure. The key insight is that expansion rate depends not on the raw number of events but on the coordinate volume subtended per event. In actualization-dense regions (galaxy clusters), each new vertex is tightly coupled to its neighbors; the metric is densely woven and resists expansion. In actualization-sparse voids, each vertex subtends a large coordinate volume because no neighboring interactions constrain the local geometry. The effective expansion rate is:

$$H_{\text{eff}}(x) \propto \frac{1}{\lambda(x)}, \tag{5}$$

where $\lambda(x)$ is the local actualization density. From this, three empirical predictions follow [Sandeman, 2026b]:

(P4) The Hubble Tension as a density-dependent signal. Early-universe probes (CMB, baryon acoustic oscillations) sample the mean λ of a dense, actively interacting young universe. Late-universe probes (Cepheid distance ladders, Type Ia supernovae) are dominated by lines of sight through low-λ voids, which expand intrinsically faster. RAGC predicts that the inferred H_0 should correlate with mean actualization density along the line of sight, with void-dominated surveys systematically returning higher values. This line-of-sight dependence is not predicted by ΛCDM (which treats H_0 as a global constant) and distinguishes RAGC from early dark energy models. No dark energy ingredient is required.

(P5) CMB power spectrum deviation from ΛCDM. If the vacuum does not gravitate, the expansion history of the universe is driven entirely by matter and radiation actualization densities, without a Λ term. The CMB angular power spectrum—particularly at low multipoles where dark energy effects are most pronounced—should deviate from ΛCDM predictions in a manner calculable from the RA-modified Einstein equation, pending the Bianchi compatibility derivation.

(P6) Cosmological constant selection in daughter spacetimes. When the actualization flux reaches $R_{\max}$, a topological severance nucleates a daughter spacetime. The Ledger Partition Theorem constrains the energy flux crossing the severance boundary; the daughter spacetime inherits an effective cosmological constant $\Lambda_{\text{eff}} \sim \Phi_{\text{boundary}}(C^*)$ determined by the local actualization flux at nucleation. This is a dynamical selection mechanism—tying Λ to the large-scale structure and matter distribution of the parent spacetime—that is sharper and more falsifiable than the anthropic landscape.

(P7) The WIMP Prohibition. A particle species with negligible electromagnetic and strong cross-sections—WIMPs, axions, sterile neutrinos—contributes negligible actualization flux and therefore negligible metric sourcing. A 100 GeV WIMP undergoes at most one on-shell interaction every 10^{18} universe lifetimes; baryonic matter in the same environment actualizes 10^{25}–10^{38} times more densely. RAGC predicts (Prediction 4) that direct detection experiments will find no WIMP signal below the neutrino floor: the absence of signal at $\sigma \lesssim 10^{-47}$ cm^2 is a derived prediction, not merely consistent. The conjectural mechanisms for flat galactic rotation curves and the Bullet Cluster lensing offset—which follow from the same actualization-density field—are developed in the companion paper RADM [Sandeman, 2026f].

3 Part III: Life as Accumulated Causal History

The foundational theorems governing biological complexity—the Assembly Index Correspondence, the DAG Markov Property, the Markov Blanket Isomorphism, the Landauer Cost of Actualization, and the Epistemic-Physical Equivalence—are proved in the companion paper RAHC [Sandeman, 2026e] as theorems about the universal causal hierarchy. The companion paper RACI [Sandeman, 2026c] develops their biological applications: the variational and computational grounding of Condition C1′, the Causal Firewall phase transition, and the evolutionary predictions. What follows is the architectural summary of both.

3.1 The Causal Firewall

Biological complexity is a problem of historical preservation. An organism is not merely *correlated* with its past; it is *constituted* by it. Its present state just *is* the irreversible sequence of actualization events that produced it. Strip away that history and you do not have a simpler organism—you have no organism at all.

This means that environments hostile to the preservation of causal history are environments hostile to the emergence of life—not statistically but *structurally*. The Causal Firewall is the physical condition under which causal history can be stably accumulated:

F1 The Quantum Darwinism redundancy threshold: the actualized state of each local system is redundantly encoded in at least $R_{\min} \sim \mathcal{O}(10)$ independent environmental fragments [Zurek, 2009].

F2 The non-relativistic condition: the Unruh temperature $T_U = \hbar a/(2\pi c k_B) \ll T_{\text{env}}$, protecting the classical record from Unruh-like disruption.

It may seem paradoxical that energetically rich, highly relativistic environments are hostile to life. Should not more interactions mean more actualization and more causal structure? The resolution is precise: the relevant quantity is not the rate of interactions but whether those interactions accumulate a *universally agreed-upon causal ledger*. In highly relativistic regimes, frame-dependence of simultaneity and the proliferation of causal horizons mean that different regions cannot maintain a consensus on the time-ordering of events. The ledger fragments. A molecular assembly process requiring a specific sequential ordering of N bond-forming events cannot be completed if different parts of the system are operating from incompatible causal histories.

3.2 The Four Empirical Predictions of RACI

Four predictions follow from the Causal Firewall and the Assembly Index framework [Sandeman, 2026c]:

(P7) The distribution of life in the universe. Life is found precisely and only in environments satisfying F1 and F2—non-relativistic, matter-dense but not radiation-dominated regimes with sufficient thermodynamic flux to drive far-from-equilibrium chemistry and sufficient causal stability for complex molecular lineages to accumulate. This is the regime of planetary surfaces around stable main-sequence stars—not because they are chemically convenient, but because they are the only environments where the causal ledger is preserved over the timescales required. The boundary of the life-permitting regime is set by the measurable condition $T_U \ll T_{\text{env}}$—a physically derived habitability criterion more fundamental than liquid water or temperature ranges, with direct implications for astrobiology and SETI.

(P8) Natural selection optimizes causal assembly depth. England's dissipative adaptation bound [England, 2013] gives replication rate $\zeta \leq (k_B T/\hbar) \cdot 2^{-A_{\text{RA}}(M)} \cdot [\text{driving term}]$. Selection for replication speed is therefore exponentially sensitive to the causal depth $A_{\text{RA}}(M)$ of the critical molecular assembly. This predicts that metabolic networks and biosynthetic pathways should exhibit

systematic pressure toward minimal Assembly Index values independently of thermodynamic cost—a prediction testable by comparing A_{RA} across lineages using Assembly Theory measurement tools [Murray et al., 2023].

(P9) The physical character of the first replicators. Applied at the origin of life, Prediction P8 yields a specific constraint: the first self-replicating systems were the causally shallowest molecules consistent with self-replication. Not the thermodynamically cheapest, not the most chemically abundant—the causally shallowest. This is a constraint on origin-of-life models: candidate first replicators should be evaluated for their A_{RA} value, not only for chemical and energetic plausibility.

(P10) The categorical impossibility of the Boltzmann Brain. Standard statistical cosmology treats the Boltzmann Brain as improbable but not impossible: given infinite time, any configuration fluctuates into existence. Relational Actualism replaces the probabilistic account with a categorical one. A brain's complexity is $A_{\mathrm{RA}}(\text{brain}) \sim 10^{25}$ or more irreversible actualization events, each individually required, each leaving a permanent causal record, none skippable. The thermodynamic steering cost bound (Theorem 5 below) establishes that a thermal fluctuation cannot fake this history—it would have to *perform* every step at full thermodynamic cost. There is no causal shortcut. In standard statistical mechanics, Boltzmann Brain improbability is quantitative: suppressed by e^{-S}, overcome given infinite time. In RA, the impossibility is categorical: no fluctuation path of the required *type* exists. A thermal fluctuation produces a momentary departure from equilibrium and returns; it cannot produce an irreversible causal record 10^{25} layers deep. The universe we observe, in which observers are embedded in a ~ 13.8 billion year causal history, is exactly what RA requires.

3.3 The Assembly Index Correspondence (Theorem 3)

The Actualism Assembly Index $A_{\mathrm{RA}}(M)$ is defined as the depth of the molecular subgraph G_M—the length of the longest directed path in the induced subgraph of all actualization events involved in assembling molecule M. This is the minimum number of sequential irreversible steps required to produce M from elementary constituents within the causal DAG.

Theorem 3 (Assembly Index Correspondence — RAHC Theorem 1 [Sandeman, 2026e]; biological application in RACI [Sandeman, 2026c])**.** *Under Condition C1′—that each JOIN operation in Assembly Theory's construction program corresponds to at least one irreversible actualization event (operationally defined as an entropy-producing on-shell energy transfer not reversed by subsequent unitary evolution)—*$A_{\mathrm{RA}}(M) \geq A(M)$*, with equality at minimal assembly.*

Condition C1′ has been established by a two-part argument. *The variational guarantee:* in the Born-Oppenheimer approximation, the variational principle dictates that the bound state energy is strictly lower than the sum of fragment energies for any stable covalent bond ($D_e > 0$). The system releases energy $|\Delta E| = D_e$ upon bond formation; to satisfy the Local Ledger Condition, this energy must be dissipated irreversibly into the environment—an actualization event under C1′. In isolated gas-phase reactions this corresponds to on-shell boson emission; in condensed-phase polymer chemistry, dissipation proceeds through many-body vibrational redistribution and solvent coupling, which under the C1′ coarse-graining rule (any entropy-producing interaction that increases $S(\rho\|\sigma_0)$ irreversibly) constitutes *more* actualization events per JOIN, not fewer. *Computational confirmation:* a B3LYP/6-311+G* density functional survey across the five primary bond types (C-C, C-N, C-O, S-S, P-O) confirms $\Delta E < 0$ in every case, with values consistent with experimental bond dissociation energies. Theorem 3 therefore holds across the full chemical alphabet. The inequality $A_{\mathrm{RA}} \geq A$ means the causal DAG records all causal history; natural selection drives $A_{\mathrm{RA}} \to A$.

3.4 The Markov Blanket Isomorphism (Theorems 4–5, Lean-Verified)

Every organism maintains an internal generative model of the external causal ledger. The physical boundary of that model is its Markov blanket: the minimal set of states that renders the organism's

internal states conditionally independent of the rest of the world. Relational Actualism derives this structure from the causal geometry rather than assuming it:

Theorem 4 (Markov Blanket Isomorphism — RAHC Theorems 2–3 [Sandeman, 2026e]; **Lean-verified core**). *In the RA causal DAG, the Markov blanket of an organism is formally isomorphic to the boundary of the organism's past light cone. Sensory states correspond to incoming boundary vertices; active states correspond to outgoing boundary vertices; the conditional independence structure is preserved under this identification.*

The Markov Blanket Shielding Theorem—the discrete graph-theoretic foundation of this result—has been formally verified in Lean 4/Mathlib. This connects Friston's Active Inference framework [Friston, 2010] to the physical process of DAG synchronization: minimizing variational free energy corresponds physically to executing the sequence of actualization events that synchronize the organism's internal DAG subgraph with the observed boundary state. The organism is not modeling the world; it is building a physical replica of the relevant fragment of the causal ledger.

Theorem 5 (Epistemic-Physical Bound — RAHC Theorem 5 [Sandeman, 2026e]; **Lean-verified**). *The thermodynamic cost of any forced actualization sequence π satisfies:*

$$W_{\text{steer}}(\pi) \;\geq\; k_B T \cdot \Delta F[q], \tag{6}$$

where $\Delta F[q]$ is the reduction in variational free energy achieved by executing π.

The organism cannot reduce its uncertainty about the world for free. Every bit of predictive accuracy about the causal ledger costs at least $k_B T \ln 2$ of free energy. Intelligence is thermodynamically expensive by necessity, not by accident.

Together, Theorems 3, 4, and 5 demonstrate that natural selection is selection for organisms that acquire predictive accuracy about the external causal ledger at minimum thermodynamic cost, via the shallowest causal assembly pathways, operating at the edge of chaos. All of the biological framework's empirical predictions (P7–P10) are consequences of this single optimization.

4 Part IV: The Information-Theoretic Turn

The first three parts develop RA at the quantum foundational, gravitational, and biological scales. Part IV extends the framework to a fourth domain: quantum information theory and its most practically consequential application, quantum computing. The full treatment appears in the companion paper RAQI [Sandeman, 2026d]; what follows is the architectural summary.

4.1 The Kinematic Snap as a Physical Decoherence Criterion

The Kinematic Snap provides what quantum computing has lacked: a precise physical criterion for when a quantum system irreversibly decoheres. Decoherence occurs when the system exchanges on-shell bosons with its environment. The actualization timescale is $\tau_{\text{act}} \sim \hbar/\Delta E_{\text{env}}$ (Prediction P2), where ΔE_{env} is the energy available in the environment for on-shell exchange. This is not a phenomenological parameter but a kinematic one.

The engineering implication is immediate: the correct target for decoherence suppression is ΔE_{env}, not coupling strength alone. Virtual, off-shell interactions—which mediate van der Waals forces, the Casimir effect, and similar phenomena—produce no DAG vertices and are categorically decoherence-immune. An interaction regime that remains purely off-shell is not merely weakly decohering; it is qualitatively protected from actualization.

4.2 The Kinematic Coherence Bound

For an N-qubit logical state, the logical information is distributed non-locally across all N qubits. Each qubit is an independent kinematic site at which actualization can occur; loss of any one

destroys the logical state. The logical decoherence rate is $N \cdot \Gamma_1$, giving logical coherence time $\tau_{\text{act}}^{(N)} = \hbar/(N \cdot \Delta E_{\text{env}})$. The maximum circuit depth before logical decoherence is the *Kinematic Coherence Bound* (KCB):

$$D_{\max} = \frac{\eta}{N}, \qquad \eta := \frac{\hbar f_{\text{gate}}}{\Delta E_{\text{env}}}, \tag{7}$$

where η is the single-qubit quality factor—the RA physical interpretation of the standard T_2/t_{gate} figure of merit. Circuit depth scales strictly as $1/N$: one cannot increase qubit count and circuit depth simultaneously without improving η. The fault-tolerance threshold condition then yields a maximum fault-tolerant array size at fixed single-qubit quality:

$$N_{\max} = \eta \cdot p_{\text{th}}. \tag{8}$$

Standard quantum error correction theory holds that fault-tolerant computation is scalable without bound once individual error rates fall below threshold. The KCB adds a physical constraint: the logical error rate grows with N, so there is a maximum array size at fixed η. For contemporary superconducting qubits ($\eta \sim 10^3$–10^4, $p_{\text{th}} \sim 10^{-2}$), this gives $N_{\max} \sim 10$–100 logical qubits—consistent with the empirical observation that scaling superconducting arrays requires continuous improvement of η alongside qubit count.

4.3 Quantum Thermodynamics Grounded in the Causal DAG

The Kinematic Snap resolves the foundational gap in quantum thermodynamics: the absence of a precise physical account of when a quantum process becomes thermodynamically irreversible.

Landauer's principle acquires a mechanical derivation. Each irreversible actualization event dissipates $W_{\text{act}} \geq \Delta q \cdot k_B T \ln 2$ (RACI Theorem 4 [Sandeman, 2026c]). RA additionally specifies the mechanism (on-shell boson emission) and a minimum timescale $\tau_{\min} \sim \hbar/\Delta E_{\text{env}}$—a joint energy-time Landauer bound stronger than the standard energy-only statement.

The Maxwell's demon paradox dissolves without information-theoretic postulates. The demon's measurement *is* an on-shell actualization event: the demon's sensory apparatus exchanges on-shell bosons with the gas molecule to determine its velocity. The thermodynamic cost is paid at measurement, not at erasure. There is no temporal gap between information acquisition and physical cost because in RA they are the same event.

The arrow of time is structural, not statistical. Each on-shell exchange writes a permanent vertex to the growing causal DAG. The past is fixed because it has been written; the future is open because it has not. This is not a consequence of the second law or of initial conditions—it is a consequence of the ontological commitment that actualization is irreversible.

4.4 The Quantum Relative Entropy Connection

The quantum relative entropy $S(\rho \| \sigma_0)$—central to both quantum information theory and the RA actualization criterion—is frame-independent by the compiled Lean 4 theorem `frame_independence` in `RA_AQFT_Proofs_v10.lean`, proved using the CFC unitary-conjugation lemma `Matrix.cfc_conj_unitary` from Lean-QuantumInfo [Meiburg et al., 2025]. An actualization event is a strict irreversible increase in $S(\rho \| \sigma_0)$; the data processing inequality bounds the information lost per actualization step.

The Generalized Quantum Stein's Lemma [Meiburg et al., 2025]—the most technically demanding theorem in physics with a computer-verified proof to date—gives an operational interpretation to the actualization criterion: $S(\rho \| \sigma_0)$ quantifies the distinguishability of the current state from the vacuum, and an actualization event is the irreversible establishment of that distinguishability beyond the quantum hypothesis testing threshold. The mathematical infrastructure underlying RA's AQFT proof program is the same infrastructure the quantum information community uses to prove its deepest theorems.

5 Part V: The Hierarchy of Complexity

Parts I–IV establish RA at four scales: the quantum foundational, the gravitational-cosmological, the biological, and the information-theoretic. Part V addresses the question that these four parts together raise but do not fully answer: *what is the physical relationship between these scales?* What generates the hierarchy from on-shell bosons to molecules to cells to minds, and why does it have the specific structure it does?

Part V also clarifies the division of labour between RACI and RAHC. RAHC [Sandeman, 2026e] is the framework paper: it proves the five foundational theorems as results about the universal causal hierarchy and establishes the recursive coarse-graining operators (Π_n) that generate every tier of physical complexity. RACI [Sandeman, 2026c] is the application paper: it takes those theorems as given and develops their biological content—the variational and computational grounding of Condition C1′, the Causal Firewall, and the evolutionary predictions. The two papers are complementary rather than overlapping.

The companion paper RAHC [Sandeman, 2026e] provides a unified formal answer. The full technical treatment appears there; what follows is the architectural summary.

5.1 The Problem of Emergence

Standard accounts of emergence face a dilemma. Either emergence is purely epistemic—a description convenient for agents who cannot track 10^{23} degrees of freedom—in which case it has no physical reality; or it is ontologically real, in which case it requires a physical mechanism that generates the higher-level properties from the lower-level ones. Most accounts choose the first horn and call it "weak emergence." RA takes the second horn seriously.

The key insight of Algorithmic Causal Dynamics (ACD), developed in RAHC [Sandeman, 2026e], is that the growing causal DAG generates a natural definition of emergence that is neither epistemic nor mysterious. It is topological.

5.2 Emergence as Topological Closure

A subgraph $S \subseteq \mathscr{G}_{n-1}$ exhibits *irreducible emergence* at scale n when it achieves sufficient causal closure that:

(i) its Markov blanket $\mathrm{MB}(S)$ is stable under the growth dynamics of $\mathscr{G}_{n-1}$; and
(ii) the macroscopic causal edges $\mathscr{E}_n$ incident on the coarse-grained node cannot be recovered from any strict sub-decomposition of S at scale $n-1$.

This is not an epistemic claim. The macroscopic edges $\mathscr{E}_n$ are physically real causal relationships that exist *only* at the coarse-grained level because they depend on the collective organisation of the subgraph. Reducing a biological organism to its constituent atoms does not reveal these edges; it destroys them. They are structural features of a particular level of causal closure, not properties of individual particles.

5.3 The Recursive Coarse-Graining Hierarchy

ACD formalises emergence via recursive coarse-graining operators (Π_n). The projection $\Pi_n : \mathscr{G}_{n-1} \to \mathscr{G}_n$ groups micro-histories of $\mathscr{G}_{n-1}$ into macroscopic equivalence classes: two micro-histories are equivalent if they produce indistinguishable probability distributions over future macro-observables at scale n. The boundary of each equivalence class is determined not by convention but by the causal topology of the underlying graph—specifically, by the Markov blanket isomorphism (RACI Theorem 2 [Sandeman, 2026c]).

Applying Π_n recursively generates the following hierarchy:

Level 0 ($\mathscr{G}_0$) The fundamental actualization events: on-shell boson exchanges satisfying $p^\mu p_\mu = m^2$. Established in Parts I and II.

Level 1 ($\Pi_1 \to \mathscr{G}_1$, ***Chemistry***) Stable covalent structures whose internal actualization history is screened by a molecular Markov blanket. Established in Part III via the Assembly Index Correspondence.

Level 2 ($\Pi_2 \to \mathscr{G}_2$, ***Cells***) Autopoietic systems maintaining a stable Markov blanket through active metabolic regulation at the edge of chaos ($\sigma \approx 1$).

Level 3 ($\Pi_3 \to \mathscr{G}_3$, ***Organisms***) Networks of cells achieving organismal-level causal closure and interacting as effective nodes in the macroscopic causal graph.

Level 4 ($\Pi_4 \to \mathscr{G}_4$, ***Agentic Intelligence***) Organisms that develop a *generative motif*: a persistent internal sub-DAG whose physical substrate is ontic (maintained by real on-shell actualization events paying the Landauer cost) but whose content is epistemic — a coarse-grained, "good enough" functional map of the environmental causal structure adequate to preserve Markov closure, not a micro-physical isomorphism. The motif can compute predictions of future environmental states before those states actualize. This is the topological criterion that distinguishes reactive autopoiesis from predictive agency.

5.4 The Mathematics of Graph Steering

At Level 4, an agent maintains an internal generative model of the environment and selects *forced actualization sequences* π that steer the growth of the global DAG toward states that preserve its Markov closure. The thermodynamic cost of any such sequence is bounded below by the Steering Cost Functional of RACI Theorem 5:

$$W_{\text{steer}}(\pi) \;\geq\; k_B T \cdot \Delta F[q], \tag{9}$$

where $\Delta F[q]$ is the reduction in variational free energy (in nats) achieved by executing π. The optimal steering policy maximises the Action Value:

$$V(\pi \mid \mathscr{G}_n) \;=\; \alpha \cdot I(\pi; \mathscr{G}_{n+1}) \;+\; \beta \cdot \mathbb{E}\big[U(\mathscr{G}_{n+1}) \mid \pi, \mathscr{G}_n\big] \;-\; \gamma \cdot W_{\text{steer}}(\pi), \tag{10}$$

where the three terms are: epistemic forcing (expected reduction in uncertainty about the future graph), pragmatic steering (expected alignment with the autopoietic objective $\sigma \approx 1$), and thermodynamic cost. The Action Value function (10) is not an imported cognitive architecture; it is derived from the RACI thermodynamic framework.

RAHC establishes the formal correspondence $V(\pi \mid \mathscr{G}_n) \approx -G(\pi) - \gamma W_{\text{steer}}(\pi)$, where $G(\pi)$ is Friston's Expected Free Energy [Friston, 2010]. The Free Energy Principle is thus the information-theoretic shadow of the RA thermodynamic reality: RA provides the physical substrate for which active inference is the functional description.

5.5 The Thermodynamic Cost of Thought

A critical consequence of ACD: the generative motif is not a disembodied computational abstraction. Its *substrate* is ontic — a physical sub-DAG maintained by real, on-shell actualization events within the agent's own graph (synaptic firings, ATP hydrolysis, ion channel dynamics in the biological case). Its *content*, however, is epistemic: a lossy, coarse-grained functional map of the environment, adequate to preserve Markov closure but not a micro-physical isomorphism. Evolution selects for adequacy, not accuracy.

"Imagination"—the internal simulation of counterfactual graph extensions—is not thermodynamically free. Each simulation step is a forced actualization sequence that pays the Landauer floor:

$$W_{\text{steer}}(\pi_{\text{sim}}) \;\geq\; k_B T \cdot \Delta F[q_{\text{sim}}]. \tag{11}$$

The human brain's consumption of approximately 20% of resting metabolic energy (Attwell and Laughlin [Attwell and Laughlin, 2001]) is a direct physical consequence: the cost of maintaining a predictive model of a complex social and physical environment is the Landauer floor for the uncertainty that model resolves. Cognition is expensive because physics requires it to be. The finite metabolic budget of any biological agent enforces a genuine trade-off between the depth of predictive simulation and the breadth of policy search — a physical account of bounded rationality.

5.6 ACD and the End of Epistemic Reductionism

The ACD framework resolves the oldest debate in the philosophy of science: whether higher levels of description are "really" lower-level physics in disguise. The answer is precise. The macroscopic edges $\mathscr{E}_n$ at Level n are physically real causal relationships that do not exist at Level $n-1$. They are generated by the coarse-graining process, which is itself constrained by the causal topology of the underlying graph. Chemistry, biology, and cognition are not reducible to physics *in the sense that the relevant causal structure is absent at the lower level*. This is not a claim about computational intractability; it is a claim about the physical reality of emergent causal edges.

The research program that ACD opens—formalising the limit theorem for the Tier 2/3 transition, connecting the coarse-graining scale to integrated information measures, quantifying the proportionality constant κ between A_{RA} and A in condensed phases—is a concrete mathematical agenda, not a philosophical position.

6 Part VI: The Mathematical Frontier

The following derivations are the core open mathematical tasks of the program. They are *derivations*—mathematical work required to connect postulates to quantitative predictions—not empirical questions. The sixteen consequences of Parts I–V stand independently. What remains is precision.

D1: The Generative Algebra of the Causal Graph. *Skeleton established.* The RA-extended Classical Sequential Growth dynamics is defined with an on-shell filter; causal invariance inherited; process algebra formulation given. **Hard wall:** the measure problem (fixing coupling constants c_k) requires QFT on causal set backgrounds. **Collaborator target:** QFT on discrete backgrounds.

D2: The DAG Continuum Limit. *Substantially resolved at the finite-dimensional level; Lean-verified.* LLC $\to$ continuity equation proved in the $\lambda \to \infty$ limit; metric reconstruction via Hawking-Malament theorem sketched. The Unruh reconciliation problem is resolved in finite dimensions: the theorems `frame_independence` and `rindler_stationarity` in `RA_AQFT_Proofs_v10.lean` establish that $S(U\rho U^\dagger \| \sigma_0) = S(\rho \| \sigma_0)$ and $\Delta S_{\mathrm{rel}} = 0$ for the Rindler KMS state (the restriction of the Minkowski vacuum to the wedge algebra, per Bisognano-Wichmann), as compiled Lean 4 theorems. The irreversible $\Delta S_{\mathrm{rel}} > 0$ that constitutes an actualization event requires a non-unitary element—the environmental trace implicit in the Local Ledger Condition—distinguishing actualization (a CPTP channel) from mere unitary evolution (which preserves relative entropy). The key lemma `log_unitary_conj`—that $\log(UMU^\dagger) = U\log(M)U^\dagger$ for Hermitian M—is a one-line corollary of `Matrix.cfc_conj_unitary` from Lean-QuantumInfo [Meiburg et al., 2025]. **Remaining hard wall:** full metric emergence from RA-CSG dynamics; extension of frame-independence to the infinite-dimensional type III_1 algebra setting via Araki's modular relative entropy [Araki, 1976]—proving $\Delta S_{\mathrm{Araki}}(\rho_{\mathrm{KMS}} \| \Omega) = 0$ on the wedge algebra $\mathscr{A}(W)$ and that $\mathscr{P}_{\mathrm{act}}$ commutes with the Tomita-Takesaki modular flow σ_s^Ω. **Collaborator target:** algebraic QFT, Tomita-Takesaki modular theory.

D3: Bianchi Compatibility for $\mathscr{P}_{\mathrm{act}}$. ***Substantially resolved in flat spacetime; curved-spacetime extension is the decisive remaining test.*** The vacuum-subtraction prescription $\langle T^{\mu\nu}_{\mathrm{RA}} \rangle := \langle T^{\mu\nu}_{\mathrm{ren}} \rangle - \langle T^{\mu\nu}_{\mathrm{ren}} \rangle|_{|0\rangle}$ satisfies $\langle 0 | T^{\mu\nu}_{\mathrm{RA}} | 0 \rangle = 0$ and reproduces Casimir physics exactly (Casimir pressure $-1.30\times$

10^{-3} Pa at $d = 1\,\mu$m, matching experiment). The conservation $\nabla_\mu \langle T^{\mu\nu}_{\mathrm{RA}} \rangle = 0$ follows by linearity and Wald axioms in flat spacetime for free fields; the flat-space benchmark verifies $\partial_\mu T^{\mu\nu} = 0$ (ordinary divergence) rather than the full covariant divergence on a curved background. The deeper resolution: the Bianchi identity is the continuum shadow of the LLC (Jacobson logic [Jacobson, 1995]). **Decisive remaining test:** Casimir apparatus in a weak gravitational field—compute $\nabla_\mu \langle T^{\mu\nu}_{\mathrm{RA}} \rangle$ (covariant divergence) under a specified Hadamard renormalization scheme and verify consistency including boundary stresses. **Remaining scope:** non-perturbative regime and full back-reaction. A related resolved question concerns the conformal trace anomaly: $\mathscr{P}_{\mathrm{act}}$ acts on the already-renormalized $\langle T^{\mu\nu}_{\mathrm{ren}} \rangle$, so the trace anomaly (arising from UV renormalization before projection) is inherited by the RA prescription and is not eliminated. This preserves consistency with Hawking radiation derivations, which depend on the trace anomaly of the renormalized theory. **Collaborator target:** renormalization theory in curved spacetime.

D4: The Graph Cut Theorem. ***Complete at the DAG level; fully verified in Lean 4/Mathlib with no `sorry` tags.*** Full proof of the Ledger Partition (RAGC Theorem 2); horizon corollary derived. All supporting lemmas—including the two sum-decomposition lemmas—are fully proved. The discrete graph-theoretic Lean file is completely `sorry`-free for all Graph Cut results. A second Lean 4 file (`RA_AQFT_Proofs_v10.lean`) formalises the finite-dimensional AQFT proof program and proves `frame_independence` and `rindler_stationarity` as compiled theorems (see D2 above). **Remaining scope:** translation to full GR boundary conditions requires Bianchi compatibility (D3); AQFT extension to type III_1 algebras (D2).

D5: The Causal Firewall Phase Transition. *Conjecture requiring a limit theorem.* Threshold equation $\lambda\,\tau_d \ell^3 = 1$ stated; England compatibility derived. **Hard wall:** elevating to a theorem requires computing τ_d explicitly from decoherence theory, proving $\sigma = 1$ corresponds to this threshold, and deriving a convergence rate (error bound in trace distance or relative entropy) for observer-dependent actualization histories converging to a shared classical history at the critical point. **Collaborator target:** open quantum systems, decoherence theory.

D6: Condition C1′ (JOIN ↔ Actualization Event). ***Established as a coarse-grained lower bound.*** Reframed from the original strict C1 (one on-shell boson per JOIN) to the robust C1′: each JOIN corresponds to at least one irreversible actualization event, where "actualization" is operationally defined as an entropy-producing interaction that increases $S(\rho\|\sigma_0)$ irreversibly under environmental trace. In isolated gas-phase reactions this is on-shell boson emission; in condensed-phase polymer chemistry it is many-body vibrational dissipation into the solvent bath—which produces *more* actualization events per JOIN, not fewer. The variational principle guarantees exothermicity ($D_e > 0$) for all stable covalent bonds; the released energy must be dissipated irreversibly, constituting a C1′ actualization event. Confirmed computationally at B3LYP/6-311+G* for all five primary bond types: C-C (-89.8 kcal/mol), C-N (-88.3), C-O (-57.2), S-S (-62.4), P-O (-84.1) — all exothermic, all consistent with experimental BDEs. **Polymer assembly:** a single abstract JOIN in polymer folding may involve $\sim 10^6$ phonon and photon transfers into the solvent. Under C1′, $A_{\mathrm{RA}}(M) \gg A(M)$ in the polymer case — the abstract Assembly Index is a strict and conservative lower bound for the immense physical causal depth required to construct biological complexity. Collective dissipation reinforces the inequality rather than threatening it. **Linear scaling:** under the thermal cascade model, $A_{\mathrm{RA}}(M) \approx \kappa \cdot A(M)$ in condensed phases where $\kappa \gg 1$ is approximately molecule-independent (energy dissipated per bond is approximately constant across bond types at fixed temperature). Selection for minimal A_{RA} is therefore simultaneously selection for minimal A; the evolutionary predictions survive intact. The constant κ is an independent experimental prediction measurable by calorimetry. **Remaining scope:** precise domain of validity for large polymer assembly where irreversibility thresholds and many-body effects require careful treatment.

The Standard Model: The Ultimate Horizon

The open derivations above represent the near-term mathematical agenda of the RA program. A longer-range horizon must be named honestly.

In its current formulation, RA remains agnostic about the origins of the three gauge forces of the Standard Model. The framework borrows electromagnetic, weak, and strong cross-sections from QFT as phenomenological inputs to the graph's transition probabilities without deriving them from the graph structure itself. A complete theory of fundamental physics cannot borrow its interaction rules; it must derive them.

The research direction is the derivation of the Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ from the pure discrete topology of the causal DAG. Particles would be stable topological motifs—localised causal knots whose characteristic actualization frequency sets their effective inertial mass. Gauge symmetries would emerge as discrete edge-routing conservation laws whose continuous limits are the Lie algebras of the Standard Model. Colour confinement would be a topological stability criterion: quarks as open subgraphs are structurally unstable and must close into baryons or mesons to achieve causal closure. This direction has precedents in Bilson-Thompson's braid model of preons and Penrose's spinor networks. The hardest specific problem is the weak force: $SU(2)$ couples exclusively to left-handed fermions, requiring the DAG to carry an intrinsic chirality whose graph-theoretic origin is not yet understood. The full treatment of this horizon appears in the RAGC Epilogue [Sandeman, 2026b].

Conclusion: One Commitment, Sixteen Consequences

The universe does not simply exist. It becomes. This is the single ontological commitment from which Relational Actualism proceeds, and the cascade of consequences is striking in both its breadth and its internal coherence.

From the commitment that actualization is primitive and physical—not observer- relative, not a subjective update, not a mathematical convenience—the following follow without ad hoc parameters:

1. **The measurement problem dissolves.** Actualization is grounded in irreversible environmental entanglement. There is no Heisenberg cut to locate; the boundary between quantum and classical is the boundary between systems that have been fully actualized by their environment and systems that have not.

2. **Quantum randomness is necessary, not accidental.** Intrinsically relational properties cannot assume definite values outside a context of causal interaction. Ontic randomness is a logical consequence of what relational properties are.

3. **The vacuum energy catastrophe disappears.** The metric is sourced only by actualized states. The vacuum contains none. $\langle 0|T^{\mu\nu}_{\mathrm{RA}}|0\rangle = 0$ structurally, not by fine-tuning.

4. **Conservation is local.** Energy-momentum is conserved at every vertex of the growing causal graph. The global version is the continuum limit of infinitely many local ledgers.

5. **Event horizons are topological severances.** Across a causal cut, each subgraph conserves independently. The energy bookkeeping version of the Hawking problem dissolves; the ledger is not lost, it is partitioned.

6. **The Hubble Tension is a geometric signal.** Void-dominated surveys return higher H_0 because voids have lower actualization density and therefore faster intrinsic expansion. The two probe families are measuring genuinely different physical regimes.

7. **The cosmological constant is selected dynamically.** Daughter spacetimes inherit Λ_{eff} from the actualization flux at their nucleation event. No anthropic landscape is required.

8. **Life is structurally constrained.** Only environments satisfying the Causal Firewall conditions can preserve the causal lineages required for biological assembly. Life is found exactly where the physics of becoming permits it.

9. **Intelligence is thermodynamically grounded.** Every bit of predictive accuracy about the external causal ledger costs at least $k_B T \ln 2$ of free energy. Natural selection optimizes causal assembly depth, edge-of-chaos operation, and faithful DAG synchronization simultaneously—all facets of a single optimization. The causal depth constraint is now computationally established: every covalent bond-forming JOIN in the assembly of biological molecules mandates an on-shell actualization event, confirmed across all five primary bond types of life (C-C, C-N, C-O, S-S, P-O) at B3LYP/6-311+G*.

10. **The Boltzmann Brain is categorically impossible.** Complexity is causal history, not configuration. A thermal fluctuation cannot fake a causal record 10^{25} layers deep. There are no shortcuts in a universe that builds its history one irreversible actualization at a time.

11. **Decoherence has a physical floor and a kinematic mechanism.** Decoherence occurs when a mediating boson crosses the on-shell threshold. The actualization timescale $\tau_{\text{act}} \sim \hbar/\Delta E_{\text{env}}$ is not a phenomenological parameter but a kinematic one. Virtual, off-shell interaction regimes are categorically decoherence-immune.

12. **Quantum computing has a kinematic scaling limit.** The Kinematic Coherence Bound establishes that circuit depth scales as $1/N$ for N-qubit logical states, and that there is a maximum fault-tolerant array size $N_{\text{max}} = \eta \cdot p_{\text{th}}$ at fixed single-qubit quality. Unlimited scaling without improving the single-qubit actualization timescale is physically impossible.

13. **Quantum thermodynamics is grounded in the causal DAG.** Landauer's principle has a mechanical derivation from the Kinematic Snap, with an additional timescale bound. The Maxwell's demon pays its thermodynamic cost at measurement, not at erasure. The arrow of time is structural—a consequence of the irreversibility of DAG inscription—rather than emergent from statistical initial conditions.

14. **Emergence is topological, not epistemic.** The recursive coarse-graining hierarchy (Π_n) generates each tier of physical complexity—chemistry, biology, cognition—as a distinct regime of causal closure. The macroscopic causal edges at each tier are physically real relationships that do not exist at the tier below: reducing a cell to its constituent atoms does not reveal them; it destroys them. Emergence is not a convenient redescription for agents with limited computational resources. It is a structural feature of the causal graph.

15. **Cognition has a physical floor.** Intelligence is the tier at which causal closure becomes sufficiently deep and recurrent that a subgraph can model the causal structure of its environment, anticipate threats to its Markov closure, and steer the growth of the DAG. The cost of this capacity—at minimum $k_B T \ln 2$ per bit of uncertainty resolved by the internal model—is not an evolutionary inefficiency. It is the Landauer floor for running a physical simulation of a complex world. The human brain's 20% resting metabolic budget is a direct thermodynamic consequence of this floor.

16. **Causally silent particles cannot gravitate.** Weakly Interacting Massive Particles (WIMPs), axions, and sterile neutrinos contribute negligible actualization flux—a WIMP undergoes at most one on-shell interaction every 10^{18} universe lifetimes—and therefore negligible metric sourcing. This is the WIMP Prohibition (RAGC Prediction 4): a derived prediction, not merely consistent with absence of signal, that direct detection experiments will find nothing below the neutrino floor. The conjectural mechanisms by which the same actualization-density field may account for

flat rotation curves and cluster lensing offsets—without invoking any new particle species—are developed in the companion paper RADM [Sandeman, 2026f]. The dark matter epicycle is not a solved problem in RA; the WIMP Prohibition is.

We are not thermodynamic anomalies rebelling against a dead statistical void. We are the natural, inevitable consequence of a universe that relentlessly writes a definite history. The mathematics of that fact—the Lean-verified Graph Cut Theorem, the computationally established Assembly Index Correspondence, the formal derivation of agency from thermodynamic first principles, the precise derivations still in progress—is the work of the research program this monograph describes.

Across six parts, this monograph has traced a single thread: the ontological commitment that actualization is primitive and physical generates, without additional assumptions, a coherent account of quantum mechanics, gravity, cosmology, life, information, and mind. The hierarchy of complexity is not an observed fact to be explained after the physics is done. It is a mathematical consequence of the physics from the beginning.

The program remains incomplete in one fundamental direction: the Standard Model. RA currently borrows the gauge forces as inputs rather than deriving them as outputs. The derivation of $U(1) \times SU(2) \times SU(3)$ from the topology of the causal DAG is the ultimate horizon of the program — the proof that the brushstrokes themselves are the inevitable algebraic symmetries of a growing graph.

The universe does not know what it is going to do. But it always knows what it has done. And that is enough.

A The Pragmatic Turn: Tractable Tools for a Discrete Ontology

A fundamental shift in physical ontology is only scientifically viable if it provides researchers with mathematically tractable tools that outperform the legacy paradigm in the regimes where it breaks down. Relational Actualism does not discard the operational successes of standard continuous theories. The Schrödinger equation remains the correct tool for calculating *potentia*—the probability distribution over possible outcomes. What RA restricts is the ontological claim that these continuous equations represent physical actuality, and it provides new tools precisely for the regimes where continuous formulations catastrophically fail: singularities, the measurement problem, and the thermodynamic emergence of complexity.

This appendix maps the specific theoretical tools RA provides to replace the intractable components of the prior paradigm. The transition is from continuous differential geometry into the mature fields of discrete mathematics, graph theory, and algorithmic complexity—fields where modern computer science is extraordinarily well-developed and where the framework's core results have been formally verified.

A.1 Replacing the Measurement Postulate: Network Thermodynamics

The legacy challenge. Standard theory relies on ad hoc postulates for wave function collapse, or on the untestable postulation of infinite universal branching (Many-Worlds), to force continuous unitary equations to yield discrete outcomes. The Heisenberg cut has no physical location; the transition from quantum to classical has no mechanism.

The RA tool: The Actualization Graph $G(V,E)$ and the Thermodynamic Ledger. RA redefines measurement not as an observer-driven epistemological update but as a strict thermodynamic threshold. Researchers can use standard Density Functional Theory (DFT) and quantum thermodynamics to map exact binding energies. If a structural transition mandates the irreversible emission of an on-shell boson to satisfy the Local Ledger Condition ($\Delta E < 0$), an actualization vertex is rigorously recorded in the causal DAG. As established in Section 3.3 and confirmed computationally

at B3LYP/6-311+G* across five bond types, this threshold is physically grounded and computable for chemically relevant reactions.

The practical advantage. Physical chemists and quantum information theorists can calculate exact, discrete causal pathways without interpretive ambiguity. Reality is computed as a Directed Acyclic Graph using standard graph theory and network flow algorithms. The measurement problem does not require a new physical postulate; it dissolves once actualization is recognized as a thermodynamic event rather than an epistemic update.

A.2 Addressing Singularities in Gravitation: Discrete Ledger Combinatorics

The legacy challenge. The continuous non-linear field equations of General Relativity ($R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$) inevitably yield infinities at high energy densities (black hole interiors, the Big Bang singularity) and predict a vacuum energy $\sim 10^{120}$ times the observed value.

The RA tool: The Local Ledger Condition and Topological Graph Cuts. In RA, spacetime is the accumulated density of the causal graph. Arbitrarily high energy density corresponds to an arbitrarily dense subgraph—not an infinity. The discrete topology does not admit singularities at the graph level; the breakdown is structural rather than analytical. The Ledger Partition Theorem (formally verified in Lean 4/Mathlib) replaces the continuous calculus of event horizons with exact boundary flux equations ($\sum_e e_{\text{charge}} = 0$), reducing the energy bookkeeping of horizons from a differential geometry problem to a network balancing problem.

Important scope note. The full replacement of Einstein's field equations requires establishing that the RA-modified stress-energy operator satisfies $\nabla^\mu \langle T^{\mu\nu}_{\text{RA}} \rangle = 0$ (Bianchi compatibility, Derivation D3) and that the discrete causal graph reproduces the GR metric in the continuum limit (Derivation D2, substantially resolved at the finite-dimensional level). These are open derivations with precise statements and partial results; they do not undermine the results established here but are required before RA constitutes a complete gravitational theory. What RA provides today is the correct discrete foundation and a proved structural suppression of the vacuum energy catastrophe.

The practical advantage. Cosmologists can work with algebraic topology and discrete combinatorics in the singularity regime, where continuous calculus fails entirely. The vacuum energy suppression is structural rather than fine-tuned. The Hubble Tension acquires a geometric mechanism: $H_{\text{eff}}(x) \propto 1/\lambda(x)$, producing a testable prediction distinguishable from ΛCDM.

A.3 Replacing Statistical Ergodicity: The RA Index and Causal Firewalls

The legacy challenge. Traditional statistical mechanics, relying on continuous phase space and ergodicity, cannot distinguish a biological organism from a random vacuum fluctuation except by appeal to improbability. The Boltzmann Brain remains logically possible. There is no physical mechanism grounding the emergence of biological autonomy.

The RA tool: The Causal Firewall and the Actualism Assembly Index A_{RA}. Astrobiologists and origin-of-life researchers gain a computable metric. $A_{\text{RA}}(M)$ quantifies the exact depth of irreversible causal history required to produce a complex structure—and this depth cannot be faked by a thermal fluctuation without performing the full thermodynamic work at every step (Theorem 5, formally verified in Lean 4/Mathlib). By identifying the Markov blanket boundary on the causal graph (Theorems 4–5, Lean-verified), researchers can mathematically isolate systems that have achieved thermodynamic autonomy, providing a biosignature framework that is physically derived rather than Earth-centric.

The practical advantage. The Boltzmann Brain is not merely improbable but categorically impossible: no fluctuation path of the required causal type exists. The distribution of life in the universe is constrained by the Causal Firewall conditions (F1 and F2), providing a habitability criterion more fundamental than liquid water or temperature ranges. These predictions are computable and testable with Assembly Theory measurement tools and astrophysical observation.

A.4 The Computational Shift

The most structurally significant upgrade is computational. Simulating quantum gravity or cosmological dynamics on a continuous manifold requires managing the approximation errors of floating-point representations of continuous space, regularization of infinities, and renormalization at every scale. These computational costs are fundamental to the formalism, not incidental.

By replacing the continuum with a discrete DAG, RA shifts the fundamental mathematics of physical reality into the domain of discrete mathematics, graph theory, and computational complexity—fields where modern computer science is extraordinarily mature. The universe, modelled as a strictly ordered, irreversible causal ledger, is natively compatible with the way modern computers process information. The Lean 4/Mathlib formal verification of the Graph Cut Theorem is not merely a proof-of-concept; it is a demonstration that the core architecture of the framework is machine-checkable at production scale.

This does not mean RA calculations are easy—fixing the generative measure of the DAG (Derivation D1) and establishing Bianchi compatibility (Derivation D3) are hard open problems. But it means that when those problems are solved, the resulting framework will be natively computable rather than requiring continuous approximation as a computational substrate.

Declarations

Funding: Not applicable. **Competing interests:** The author declares no competing interests.

References

H. Araki. Relative entropy of states of von Neumann algebras. *Publications of the Research Institute for Mathematical Sciences*, 11(3):809–833, 1976.

J. F. Sandeman. Algorithmic Causal Dynamics: A Relational Actualist Framework for Emergence and Complexity (RAHC). Preprint, 2026.

D. Clowe, M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky. A direct empirical proof of the existence of dark matter. *The Astrophysical Journal Letters*, 648:L109–L113, 2006.

F. Lelli, S. S. McGaugh, and J. M. Schombert. SPARC: Mass models for 175 disk galaxies with *Spitzer* photometry and accurate rotation curves. *The Astronomical Journal*, 152:157, 2016.

J. F. Sandeman. Relational Actualism: Dark Matter as Actualization-Density Topology (RADM). Preprint in preparation, 2026.

Sandeman, J. F. (2026a). The Relational Universe: Interaction and the Transition of the Possible to the Actual. *Foundations of Physics* (Submitted).

Sandeman, J. F. (2026b). Relational Actualism Gravity and Cosmology (RAGC). Preprint.

Sandeman, J. F. (2026c). Relational Actualism Complexity and Intelligence (RACI). Preprint.

Rovelli, C. (1996). Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8):1637–1678.

Bassi, A., Lochan, K., Satin, S., Singh, T. P., and Ulbricht, H. (2013). Models of wave-function collapse, underlying theories, and experimental tests. *Reviews of Modern Physics*, 85(2):471–527.

Diósi, L. (1989). Models for universal reduction of macroscopic quantum fluctuations. *Physical Review A*, 40(3):1165–1174.

Penrose, R. (1996). On gravity's role in quantum state reduction. *General Relativity and Gravitation*, 28(5):581–600.

Weinberg, S. (1989). The cosmological constant problem. *Reviews of Modern Physics*, 61(1):1–23.

Jacobson, T. (1995). Thermodynamics of spacetime: the Einstein equation of state. *Physical Review Letters*, 75(7):1260–1263.

Dowker, F. (2005). Causal sets and the deep structure of spacetime. *100 Years of Relativity*, pages 445–467.

Sorkin, R. D. (2003). Causal sets: Discrete gravity. *Lectures on Quantum Gravity, NATO Science Series*, pages 305–327.

Rideout, D. P. and Sorkin, R. D. (2000). Classical sequential growth dynamics for causal sets. *Physical Review D*, 61(2):024002.

Zurek, W. H. (2009). Quantum Darwinism. *Nature Physics*, 5(3):181–188.

D. Attwell and S. B. Laughlin. An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*, 21(10):1133–1145, 2001.

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138.

England, J. L. (2013). Statistical physics of self-replication. *Journal of Chemical Physics*, 139(12):121923.

Murray, A. R. G. et al. (2023). Defining and measuring the Assembly Index. arXiv:2206.02384 [q-bio.QM]. *Journal of Molecular Evolution* (in press).

Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191.

Kirchhoff, M., Parr, T., Palacios, E., Friston, K., and Kiverstein, J. (2018). The Markov blankets of life. *Journal of the Royal Society Interface*, 15(138):20170792.

Kauffman, S. A. (2000). *Investigations*. Oxford University Press.

Minev, Z. K. et al. (2019). To catch and reverse a quantum jump mid-flight. *Nature*, 570:200–204.

Pazourek, R., Nagele, S., and Burgdörfer, J. (2015). Attosecond chronoscopy of photoemission. *Reviews of Modern Physics*, 87:765–802.

Smolin, L. (2013). *Time Reborn*. Houghton Mifflin Harcourt.

Tomonaga, S. (1946). On a relativistically invariant formulation of the quantum theory of wave fields. *Progress of Theoretical Physics*, 1(2):27–42.

Schwinger, J. (1948). Quantum electrodynamics I: A covariant formulation. *Physical Review*, 74(10):1439–1461.

A. Meiburg, L. A. Lessa, and R. R. Soldati. A formalization of the generalized quantum Stein's lemma in Lean. arXiv:2510.08672, 2025. Lean-QuantumInfo: `https://github.com/Timeroot/Lean-QuantumInfo`.

J. F. Sandeman. Actualization as physical mechanism: Implications of relational actualism for quantum computing and quantum thermodynamics. Preprint, 2026.